

Known classification for Conjecture 2 of Montgomery–Smith–Semenov

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Status

This is a *literature-implied answer (full classification)* for Conjecture 2 of Montgomery–Smith and Semenov [1]. The packet records a later accessible arXiv source which states the exact classification as known and cites the underlying strict-singular embedding literature.

Source Question

The source paper is S. J. Montgomery-Smith and E. M. Semenov, *Embeddings of rearrangement invariant spaces that are not strictly singular*, arXiv:math/0007058, Positivity 4 (2000), 397–404. On PDF page 2, Conjecture 2 states:

the natural embedding $E \subset L_1$ is not strictly singular $\iff G$ embeds into E continuously,

where E is rearrangement invariant on $[0, 1]$ and G is the closure of the simple functions in the Orlicz space for $\exp(x^2) - 1$.

Supporting Identification

Hernandez, Ruiz, and Sanchiz [2], Introduction, PDF page 1, recall the following known fact for symmetric, i.e. rearrangement invariant, function spaces on finite measure spaces:

$$E(\mu) \hookrightarrow L_1(\mu) \text{ is strictly singular} \iff L_0^{\exp x^2} \text{ is not included in } E(\mu).$$

This is precisely the contrapositive form of Conjecture 2, with $L_0^{\exp x^2}$ being the Rodin–Semenov space G used in the source. The supporting paper cites the results of Hernandez–Novikov–Semenov [3] and Astashkin–Hernandez–Semenov [4].

Scope

This packet does not claim a new proof. It records that the main strict singularity classification in Conjecture 2 is already known in the later literature. It also does not address the stronger sign Conjecture 3; that printed statement is handled separately by the companion counterexample packet.

References

- [1] S. J. Montgomery-Smith and E. M. Semenov, *Embeddings of rearrangement invariant spaces that are not strictly singular*, Positivity 4 (2000), 397–404. <https://arxiv.org/abs/math/0007058>
- [2] F. L. Hernandez, C. Ruiz, and M. Sanchiz, *Disjointly strictly singular inclusions between variable Lebesgue spaces*, arXiv:2406.14175, 2024. <https://arxiv.org/abs/2406.14175>
- [3] F. L. Hernandez, S. Ya. Novikov, and E. M. Semenov, *Strictly singular embeddings between rearrangement invariant spaces*, Positivity 7 (2003), 119–124.
- [4] S. V. Astashkin, F. L. Hernandez, and E. M. Semenov, *Strictly singular inclusions of rearrangement invariant spaces and Rademacher spaces*, Studia Math. 193 (2009), 269–283.